

JavaScript (35 to 40 Days) — Class Notes

1. Introduction

Technology	Purpose
HTML	Provides the structure of a web page
CSS	Adds styling and layout to a web page
JavaScript (JS)	Makes web pages dynamic and interactive

🧠 JavaScript is used to build dynamic web applications.

2. History of JavaScript

- Introduced: 1995
 - Created by: *Brendan Eich*
 - Development time: 10 days
 - Company: *Netscape Communications*
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3. Why is it called JavaScript?

- Developed by Brendan Eich at Netscape.
 - Original project name: Mocha
 - Renamed to LiveScript
 - Finally renamed to JavaScript (for marketing reasons — to capitalize on Java's popularity).
 - Microsoft created its own version called JScript.
 - Java (by *Sun Microsystems*) and JavaScript are completely different languages.
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4. JavaScript Engines

Browser	JavaScript Engine
Google Chrome	V8
Mozilla Firefox	SpiderMonkey
Microsoft Edge	Chakra
Tor Browser	SpiderMonkey

🌱 Note:

- Netscape browsers originally supported JavaScript.

- Older Internet Explorer versions had limited or inconsistent JavaScript support.
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5. ECMAScript (ECMA)

- ECMA stands for European Computer Manufacturers Association.
 - ECMAScript is the standardized version of JavaScript.
 - It defines language rules and syntax so JavaScript works consistently across all browsers.
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6. Interpreter vs Compiler

Feature	Interpreter	Compiler
Execution	Executes line by line	Translates the entire code before running
Speed	Slower	Faster
Error Handling	Stops at the first error	Displays all errors after compilation
Output Type	Immediate output	Needs compilation step first

Example:

- `console.log(1 + 2) // 3`

- `console.log(3 + 4) // 7`
- `prnt("error1") // Error: 'prnt' is not defined`
- `console.log("hello")`

Interpreter Output:

- 3
- 7
- Error at line 3

Compiler Output:

- Error at line 3 and line 5

⚙ Note:

JavaScript is interpreted, but modern JS engines (like V8) use JIT (Just-In-Time Compilation) — a mix of interpreter + compiler for better performance.

7. Output Statements

- `console.log("Hello Developers")`
- `console.log(2 + 3) // Output: 5`

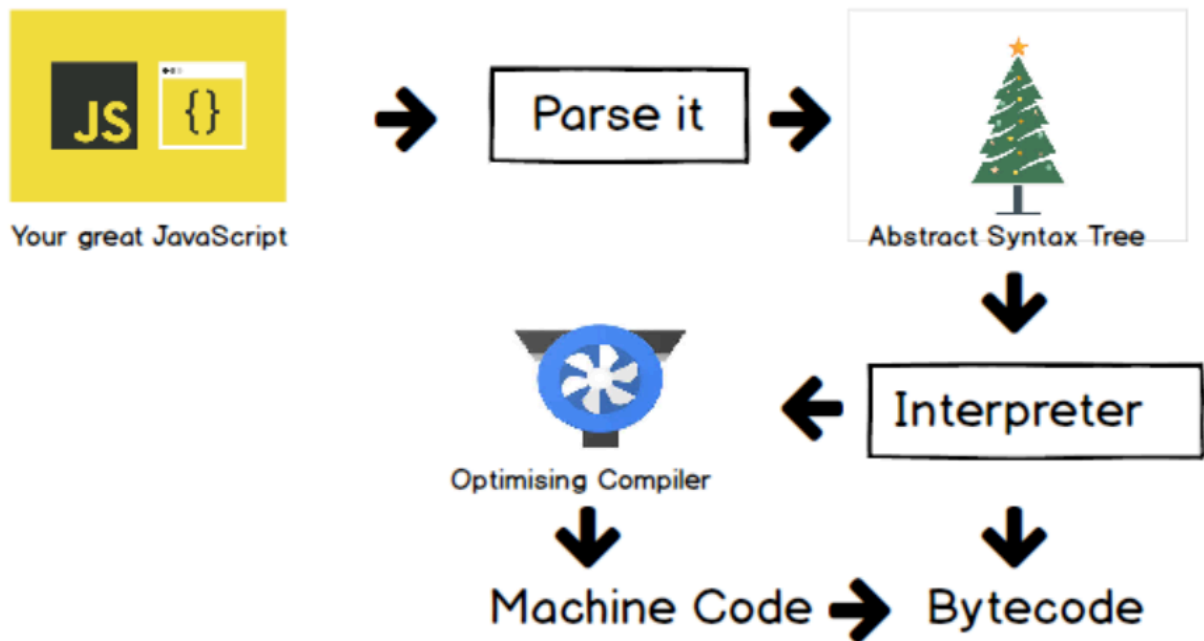
⚠ Note:

`print()` is not a valid JavaScript function — use `console.log()` instead.

8. How JavaScript Executes (Behind the Scenes)

- Source Code
- ↓
- Parsing

- ↓
- AST (Abstract Syntax Tree)
- ↓
- Interpreter
- ↓
- Bytecode
- ↓
- JIT Compiler (Optimized Machine Code)



Example (Valid Syntax):

- console
- .
- log
- (
- 2
- +
- 3
-)

9. Applications of JavaScript

JavaScript is a multi-purpose language used in many domains:

Application Type	Examples / Frameworks
Web Development	React, Angular, Vue
Backend Development	Node.js, Express.js
Mobile Applications	React Native, Ionic
Desktop Applications	Electron.js (used in VS Code, Discord)
Game Development	Phaser.js, Three.js
Machine Learning / AI	TensorFlow.js
IoT (Internet of Things)	Johnny-Five.js

10. Summary

- JavaScript was created by Brendan Eich in 1995.

- It was originally called Mocha, later LiveScript, and finally JavaScript.
- JavaScript is standardized by ECMAScript.
- It runs on different engines like V8 and SpiderMonkey.
- JavaScript uses JIT compilation for efficiency.
- It is used to create web, mobile, desktop, and gaming applications.